

Ex.no:7b	DATA VISUALIZATION USING TABLEAU (CITY DAY AIR QUALITY)

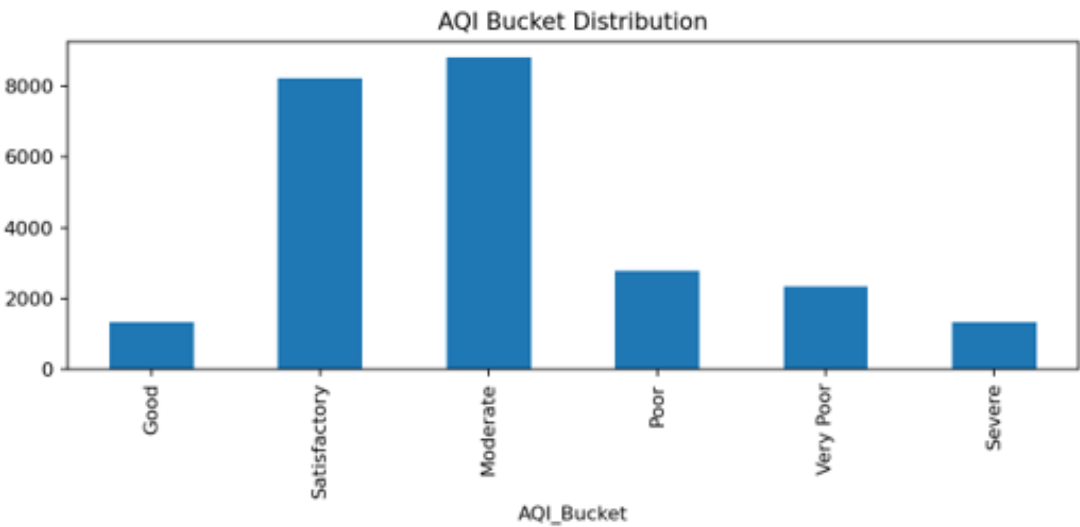
Aim:

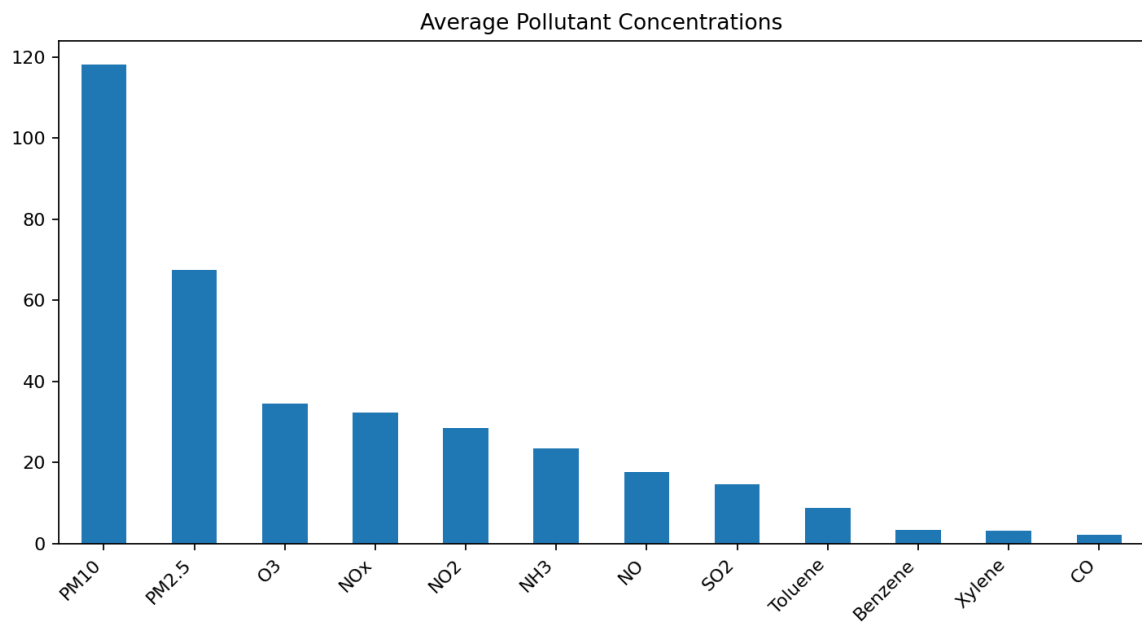
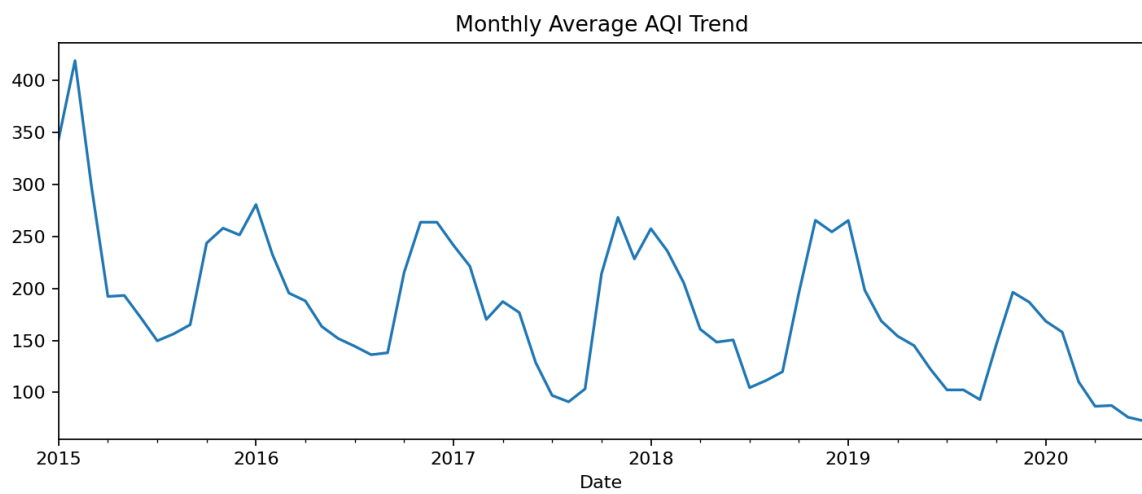
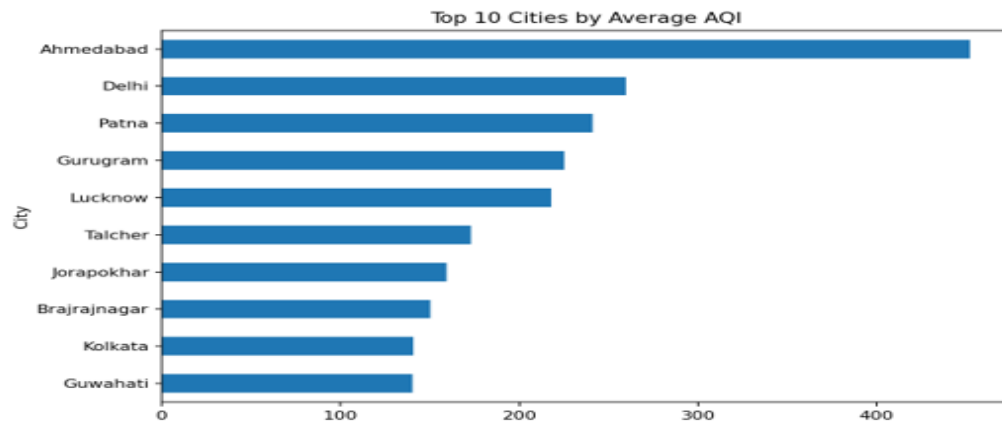
To perform data analysis, data preparation, and visual exploration on the City Day air-quality dataset (city_day.csv) in Tableau Desktop/Prep to analyze AQI levels, pollutant concentrations, city-wise air quality, and time-based trends for actionable insights.

Algorithm:

- 1. Data Ingestion & Connection:** Connect Tableau Desktop to city_day.csv as a text/CSV data source.
- 2. Verify Data Types:** Set Date as Date and verify numeric AQI and pollutant fields.
- 3. Data Cleansing & Transformation:** Handle missing values in pollutant, AQI, and categorical fields using suitable imputation methods.
- 4. Create Calculated Fields:** Create calculated fields for Average AQI, Maximum AQI, and record count.
- 5. Exploratory Visual Analysis:** Create worksheets for AQI category distribution, city-wise AQI, pollutant comparison, and monthly AQI trends.
- 6. City-wise Analysis:** Compare average AQI across cities and identify cities with higher AQI levels.
- 7. Seasonality Trends:** Plot Date against average AQI to observe changes in air quality over time.
- 8. Dashboard Assembly & Formatting:** Combine key worksheets into an interactive Tableau dashboard and apply filters such as City, Date, and AQI Bucket.

Output:





Result: